

A Comparative Evaluation of Linear Discriminant Analysis and Logistic Classification using Simulation and Real-World Data Applications

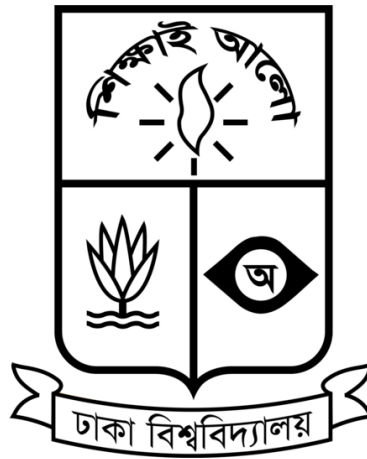
Submitted by:

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Roll: FH-182-002

Session:2024-25

4th year B.S in Applied Statistics



Supervised By:

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University of Dhaka

*A project submitted for partial fulfillment of the requirement for the Degree in B.S.(Hons.) in
Applied Statistics*

Institute of Statistical Research and Training

University of Dhaka

Dhaka-1000, Bangladesh.

March 2026

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Declaration

I, hereby, certify that the project report entitled "A Comparative Evaluation of Linear Discriminant Analysis and Logistic Classification using Simulation and Real-World Data Applications", which has been submitted as a partial requirement for the degree of B.S. (Hons.) in Applied Statistics is the result of my own research, except where otherwise acknowledged, and that this project report in whole or in part has not been submitted for an award, including higher degree to any other University or Institution.

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Session: 2021-22

Institute of Statistical Research and Training (ISRT)

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Dedicated to my

family

friends

Teachers

Acknowledgement

First, I want to express my heartfelt gratitude to the Almighty for providing me with the ability to complete this study. After that, I want to convey my sincere admiration for my respected supervisor, Dr. Mohammad Shafiqur Rahman, Professor, Institute of Statistical Research and Training (I.S.R.T.), University of Dhaka, whose tremendous effort helped me throughout my journey. This project would not have been possible without his consistent direction and continuous support. Finally, I want to thank my family, my teachers, and all my friends for always being there for me and for helping me accomplish this fine task.

Abstract

Logistic Regression and Linear discriminant Analysis are two supervised learning models that have long been used for dichotomous discrimination studies. In public health, it is important to know which of these models gives better predictions given the circumstances. A Monte Carlo Simulation has been engineered for that very reason to make the process of selecting a model more fathomable for the non statisticians given their situation and data at hand. Through repeated sampling under controlled conditions, our simulation evaluates model performance across varying sample sizes, correlations, distances, and distributional assumptions. This allows practitioners to observe how each method behaves when key assumptions—such as normality or equal covariance structures—are satisfied or violated. Performance metrics such as accuracy, sensitivity, specificity, and area under the ROC curve are compared to provide a complete understanding of predictive ability.

The results of such simulations can guide decision-making in real-world public health applications, where data quality and structure often differ from textbook assumptions.

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Chapter 1

Introduction

1.1 Purpose of the Study

Both Logistic regression and Linear Discriminant Analysis can be used to predict the probability of a specified categorical outcome using several explanatory variables. The dilemma, regarding which model between these two performs better in classification, has continued to baffle statisticians for a long time. On one hand, Logistic regression is very popular in the health sciences, since the discrete outcome could often be the presence or absence of a disease. On the other hand, LDA also performs a good discrimination in such a dichotomous situation. One thing is certain; Both models can be used to address the same type of research problem.

Despite their similar objectives, the two methods differ fundamentally in their theoretical foundations and assumptions. Logistic Regression is a probabilistic, model-based approach that directly estimates the conditional probability of class membership without requiring strong distributional assumptions for the predictors. In contrast, LDA is a generative method that models the distribution of predictors within each class and then applies Bayes' theorem to perform classification. Because of these differences, their relative performance may vary depending on factors such as sample size, class separation, multicollinearity among predictors, and violations of distributional assumptions.

In applied fields such as public health and epidemiology, selecting an appropriate classification model is not merely a theoretical concern but a practical necessity. Incorrect model choice may lead to biased risk predictions, poor classification accuracy, or misleading policy recommendations. Therefore, understanding the conditions under which each method performs optimally is essential for evidence-based decision-making. This study aims to explore and compare the predictive performance of Logistic Regression and LDA under different simulated scenarios, providing guidance for researchers in choosing the most suitable method for their specific data characteristics and research objectives.

The objective of this work is to investigate whether these two methods of analysis result similar findings in evaluating categorical health outcomes.

1.2 Objective of the Study

The main objectives are as follows:

1. Generating different data sets surrounding a particular clinical problem to cover variety of scenarios
2. Applying both LDA and logistic regression to each simulated case and obtain classification result
3. comparing model by evaluating the calculated indexes and finding the proportion of better performance
4. Applying both the models to a real world Dataset (Non-normal Scenario)
5. Discussing future prospects

1.3 Literature Review

This Comparison between logit and LDA model has been extensively scrutinized by researchers since the mid 20th century, following Fisher's introduction of discriminant analysis in 1936 ([Fisher, 1936](#)) and later on the development of logistic regression by Cox in 1958 ([Cox, 1958](#)). Naturally, Researchers began systematically evaluating their relative performance under different data conditions.

One of the Earliest papers on this topic provides us with the basis for theoretical comparison. ([Efron, 1975](#)) The key findings of this paper pointed us to the fact that Logistic regression remains efficient even when normality assumptions fail. And under normality, both methods are similar. Elsewhere, we can see that Logistic Regression is often preferred in practical use due to fewer assumptions. ([Press and Wilson, 1978](#))

But our goal mainly aligns with the recent publication where the author mainly created different scenarios to compare LDA and logistic regression under different assumption violations. ([Pohar et al., 2004](#)) The study showed that logistic regression demonstrates greater robustness when the assumptions required for Linear Discriminant Analysis, particularly multivariate normality and equality of covariance matrices, are violated. In practical datasets where these assumptions rarely

hold perfectly, logistic regression tends to maintain stable predictive performance. When the assumptions of discriminant analysis are satisfied, especially when predictor variables follow a multivariate normal distribution with homogeneous covariance structures across groups, LDA performs comparably to logistic regression and may even achieve slightly higher classification accuracy.

The authors(Pohar et al., 2004) observed that LDA can perform well in smaller samples when distributional assumptions are met, whereas logistic regression benefits more from larger sample sizes because it relies on maximum likelihood estimation without strict distributional assumptions. Logistic regression was found to provide more stable parameter estimates in situations involving skewed predictors, unequal variances, or outliers, conditions that can substantially degrade the performance of discriminant analysis. A major contribution of the study was the conclusion that neither method is universally superior.

Next we can focus on comparison papers that utilized real world datasets for classification tasks involving binary outcomes(Liong and Foo, 2013). Both LDA and logistic regression models were fitted to the same datasets (multiple) to ensure a fair comparison. The models were later compared based on classification accuracy i.e. miss classification rate and predictive capability. In the end, results from both models were analyzed to confirm which performed better. Ultimately, the study found that both LDA and logistic regression produced similar classification accuracy in many cases, particularly when the dataset approximately satisfied LDA assumptions. Logistic regression was highlighted as more interpretable by the authors. They concluded that Logistic regression demonstrated greater flexibility because it does not require strict assumptions; but when assumptions of multivariate normality and equal covariance matrices were reasonably satisfied, LDA performed very well and sometimes slightly better than the later.

The paper which focuses on comparing classification error rates between Linear Discriminant Analysis (LDA) and logistic regression in the two-group classification case (binary outcome) is noteworthy (Lei and Koehly, 2003). The goal was to determine under what conditions one method produces lower misclassification error than the other. The authors used a Monte Carlo simulation study and created a general outline which has now become our roadmap.They simulated datasets under controlled conditions, generated predictor variables from both multivariate normal and non-normal distribution(to violate LDA assumptions), varied several conditions systematically for instances sample size, equality and inequality of covariance matrices, degree of non-normality and group separation. Afterwards, both LDA and logistic regression were fitted to each simulated dataset. Misclassification error rate was used as the primary comparison metric. Their findings are very similar to what was already known and they noted that with smaller sample sizes, LDA sometimes performed competitively or slightly better when assumptions held. But with larger samples, logistic regression became more stable and often preferable. When group separation was large, both methods performed similarly because classification became easier regardless of the

method.

But our study will remain incomplete if we do not compare the performance of these two models with a real world dataset. One such study where real world epidemiological datasets has been utilized is of interest to our study.([Panagiotakos, 2006](#)) The dataset used by the authors consists of 2,172 patients (1,649 men and 523 women) admitted for Acute Coronary Syndrome (ACS) between October 2003 and September 2004. Both the models identified the same set of variables as the most important predictors of mortality (CK-MB levels, age, systolic blood pressure, sex, and diabetes). Even though the data violated a key assumption of LDA(the covariance matrices were not equal); LDA still performed well and agreed with Logistic Regression. Ultimately, when comparing the ROC curves, they found no statistical difference between the two models. It is noteworthy that they used a bootstrap simulation (resampling the data 1000 times) to confirm that the results were stable and not just due to chance.

Another challenge encountered when working with real-world datasets is selecting significant independent variables that adequately explain the response variable. For this study, we focus on one of the most pressing areas of public health research: hypertension among the surveyed population. We have looked into several studies mainly focusing on hypertension utilizing the Bangladesh Demographic and Health Survey(BDHS) dataset. Almost all the studies had a specific way of defining the response variable and selected a specific set of explanatory variables ([Chowdhury et al., 2016](#); [Al Kibria et al., 2021](#); [Khan et al., 2021](#); [Ahmed et al., 2019](#)). Most notably the authors have always used variables like age, sex, body mass index, wealth index, diabetes among the respondents and have always set a specific medically determined criteria to determine the presence of hypertension.

Taken together, the existing body of literature consistently suggests that neither Linear Discriminant Analysis nor logistic regression is uniformly superior across all data conditions. Rather, their relative performance depends critically on distributional assumptions, covariance structure, sample size, and degree of group separation. Early theoretical contributions ([Efron, 1975](#)) established the foundation for understanding their asymptotic efficiency, while applied and simulation-based studies ([Press and Wilson, 1978](#); [Lei and Koehly, 2003](#); [Pohar et al., 2004](#); [Liong and Foo, 2013](#)) further clarified the practical implications of assumption violations and sample characteristics.

Despite the extensive scrutiny devoted to this comparison, the evidence remains context-dependent, with results often varying according to experimental design and performance metrics used. This suggests that the comparison between LDA and logistic regression is not a settled question, but rather one that requires careful evaluation under clearly defined and controlled conditions.

1.4 Organization of the Study

This study is organized into five chapters. An overview of the purpose of the study, a comprehensive literature review, and a description of the objectives of this study are provided in Chapter 1. The methodology employed in this study are explained in Chapter 2. Chapter 3 is informative on the real world Dataset and variables used. The next Chapter, which is Chapter 4, elaborates on the analysis and findings of the study. Last but not the least, Chapter 5 includes discussions and conclusions drawn from the study, and emphasizes on the limitations of the study and future undertakings for other studies.

1.5 Expectation

The goal of our study has been extensively investigated before in various simulation studies and suitable conclusions have been reached. The systematic approach and methodology used in the previous mentioned papers have been adopted in our work with slight disparity to analyze the predictive and discriminatory capability of the models in questions. This work of ours does not intend to propose new theoretical developments, rather the primary aim was always to apply, reproduce, and critically assess existing methodologies in order to gain practical insight into their behavior under varying data conditions. In the end(in conclusion) we have tried to point out the discrepancies of our work and how better approaches are already available and should be utilized to reach an ultimate decision for any and all possible scenarios(datasets).

Chapter 2

Methodology

2.1 Receiver Operating Curve

The Receiver Operating Characteristic (ROC) curve is an essential tool for evaluating and comparing the performance of binary classification models. It can provide a visual representation of a model's discriminatory power across all possible classification thresholds, rather than being limited to a single cut-off point.

The curve is constructed by plotting the True Positive Rate (Sensitivity) against the False Positive Rate (1 - Specificity) for every possible threshold value. The True Positive Rate measures the proportion of actual positive cases that are correctly identified by the model. The False Positive Rate represents the proportion of actual negative cases that are incorrectly classified as positive.

A model with no discriminatory power, essentially random guessing, would produce an ROC curve that follows the diagonal line from (0,0) to (1,1). Conversely, a perfect classifier would have a curve that passes through the top-left corner (0,1), indicating 100% sensitivity and 100% specificity.

2.2 Performance Evaluation Indices

To evaluate and compare the predictive performance of Logistic Regression and Linear Discriminant Analysis (LDA), three discrimination indices were used: the C-index, Q-index, and B-index. These indices are derived from the Area Under the Receiver Operating Characteristic Curve (AUC) and provide different perspectives on model accuracy. All the indices below uses Y which is the observed outcome and P which means the predicted probability.

2.2.1 C-index(AUC)

The C-index measures the probability that, for a randomly selected pair of individuals, the model assigns a higher predicted probability to the individual with the event than to the individual without the event. The value of the C-index ranges from 0.5 (No discrimination i.e. random guessing) to 1 (Perfect discrimination),

$$\begin{aligned} C &= \frac{\text{Concordant Pairs} + 0.5 \times \text{Tied Pairs}}{\text{Total Pairs}} \\ &= \sum_{\substack{i=1 \\ Y_i=0}}^n \sum_{\substack{j=1 \\ Y_j=1}}^n \left[I(P_j > P_i) + \frac{1}{2} I(P_j = P_i) \right] / n_0 n_1. \end{aligned}$$

Here, Concordant Pair: Positive instance has a higher probability than the negative instance. Tied Pair: Positive and negative instances have the same probability. n_0 denotes the number of observation belonging to class(0) and n_1 for class(1). A higher C-index indicates better predictive ability of the model.

2.2.2 Q-index

The Q index is a performance measure derived from the logarithmic score. It evaluates the quality of predicted probabilities, not just classifications. It is closely related to cross-entropy and log-likelihood, and unlike classification error rate, it evaluates the quality of predicted probabilities rather than discrete classifications; given by,

$$Q = 1 + \frac{1}{n} \sum [Y \log_2(P) + (1 - Y) \log_2(1 - P)].$$

A value of $Q = 1$ corresponds to perfect predictions, while $Q = 0$ indicates no predictive information, equivalent to random guessing. Conversely, a value of $Q < 0$ signifies performance worse than random.

2.2.3 B-index

Brier Score is a scoring rule that measures the reliability of probabilistic predictions. It does so by computing the average squared difference between predicted probabilities and the actual outcomes across all possible classes (R represent the number of classes),

$$B = \frac{1}{N} \sum_{t=1}^N \sum_{i=1}^R (P_{ti} - Y_{ti})^2.$$

Lower values indicate better predictive performance. We have used $1 - B$. So the higher the value the better.

2.3 Monte Carlo Simulation

Monte Carlo simulation is a computational technique that repeatedly generates data under controlled conditions in order to assess the effectiveness of statistical methods. The method generates a large number of artificial samples from a specified data-generating process rather than depending on a single dataset. Each generated dataset is then fitted to the models under investigation, and their performance metrics are noted. Stable and trustworthy estimates of model performance can be obtained by averaging the outcomes over a large number of repetitions.

As replication increases, the Monte Carlo estimate converges to the true expected value of the performance measure. This approach reduces the influence of random sampling variability and provides a more comprehensive comparison of the competing classification methods.

The use of Monte Carlo simulation allows for systematic investigation of model behavior under different sample sizes, class distributions, and distributional assumptions. Therefore, it provides a robust framework for evaluating and comparing the predictive accuracy of Logistic Regression and Linear Discriminant Analysis.

2.4 Bootstrapping

Bootstrapping is a resampling technique used to estimate the sampling distribution of a statistic by repeatedly sampling with replacement from the original data. The non-parametric bootstrap is the most commonly used technique. In this approach, each bootstrap sample is created by randomly drawing n observations with replacement from the original dataset of size n . Because sampling is with replacement, any given observation may appear multiple times or not at all in a particular bootstrap sample.

This method makes no assumptions about the underlying distribution and is applicable to virtually any statistic, from simple means to complex regression coefficients. Its flexibility and simplicity have made it the workhorse of modern resampling methods.

2.5 Logistic Regression Model

Logistic regression is a supervised learning technique used to model the probability of a binary outcome. Unlike linear regression, which predicts a continuous value, logistic regression predicts the probability that an observation belongs to a specific class (typically coded as 1 versus 0). It achieves this by applying the logistic (sigmoid) function to a linear combination of predictor variables, thereby mapping the output to a probability bounded between 0 and 1.

Formally, for a given observation i , the outcome Y_i is assumed to follow a Bernoulli distribution with probability p_i ,

$$Y_i \mid X_i \sim \text{Bernoulli}(p_i),$$

where $E[Y_i \mid \mathbf{X}_i] = p_i = \Pr(Y_i = 1 \mid \mathbf{X}_i)$.

The core of the model lies in linking this probability p_i to a linear predictor function, $\boldsymbol{\beta}^T \mathbf{X}_i$, which is a weighted sum of the p predictor variables:

$$\boldsymbol{\beta}^T \mathbf{X}_i = \beta_0 + \beta_1 x_{1,i} + \cdots + \beta_p x_{p,i} .$$

To ensure the model's output is a probability, logistic regression uses the logit link function. This function transforms the probability p_i into the log-odds, $\ln\left(\frac{p_i}{1-p_i}\right)$, which has a range of $(-\infty, +\infty)$ and can therefore be modeled as a linear function of the predictors. The model is formally expressed as:

$$\text{logit}(p_i) = \ln\left(\frac{p_i}{1-p_i}\right) = \boldsymbol{\beta}^T \mathbf{X}_i .$$

By solving for p_i , we obtain the core logistic regression equation, which directly gives the probability of the outcome:

$$p_i = \Pr(Y_i = 1 \mid \mathbf{X}_i) = \frac{1}{1 + e^{-\boldsymbol{\beta}^T \mathbf{X}_i}} = \frac{e^{\boldsymbol{\beta}^T \mathbf{X}_i}}{1 + e^{\boldsymbol{\beta}^T \mathbf{X}_i}} .$$

This S-shaped function ensures that any value of the linear predictor $\boldsymbol{\beta}^T \mathbf{X}_i$ is mapped to a probability value between 0 and 1.

2.5.1 Assumptions for logit model

Many of the fundamental assumptions of linear regression models and generic linear models based on ordinary least squares algorithms are not made by logistic regression models, especially with regard to linearity, normalcy, homoscedasticity, and measurement level. However, some other assumptions still apply.

1. Nature of the outcome variable: In binary logistic regression, the dependent variable must have two categories.
2. Independence of observations: The method assumes that all observations are independent.
3. Absence of multicollinearity: Logistic regression assumes that the predictor variables are not highly correlated with one another.

4. Linearity in the logit: Although logistic regression does not require a direct linear relationship between the dependent and independent variables, it assumes that the independent variables have a linear relationship with the log odds (logit) of the outcome.

2.6 Linear discriminant Analysis

Linear Discriminant Analysis (LDA) is a dimensionality reduction and classification technique that identifies a linear combination of features which best separates two or more classes.

For a two-class problem, LDA assumes that the predictor variables $\mathbf{X} = (x_1, \dots, x_p)$ follow a multivariate normal distribution within each class k , with class-specific mean vectors $\boldsymbol{\mu}_k$ but a common covariance matrix $\boldsymbol{\Sigma}$. The goal is to find a linear discriminant function, $f(\mathbf{X}) = \boldsymbol{\beta}^T \mathbf{X}$, that maximizes the separation between the classes.

Fisher's criterion formalizes this by finding the vector $\boldsymbol{\beta}$ that maximizes the ratio of the between-class variance to the within-class variance,

$$\boldsymbol{\beta} \propto \boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1) .$$

Using Bayes' theorem, the probability that an observation $\mathbf{X}$ belongs to class k is given by,

$$\Pr(Y = k \mid X = x) = \frac{\pi_k f_k(x)}{\sum_{j=1}^K \pi_j f_j(x)} .$$

where π_k is the prior probability of class k , and $f_k(\mathbf{x})$ is the multivariate normal density for class k . Taking the logarithm and simplifying under the assumption of equal covariance matrices yields the linear discriminant function:

$$\delta_k(x) = x^T \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}_k - \frac{1}{2} \boldsymbol{\mu}_k^T \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}_k + \log \pi_k .$$

An observation $\mathbf{x}$ is assigned to the class k for which $\delta_k(\mathbf{x})$ is largest. The decision boundary between any two classes is linear in $\mathbf{x}$, giving the method its name.

2.6.1 Assumptions for LDA

The underlying requirements of discriminant analysis are similar to those associated with multivariate analysis of variance (MANOVA). This method is particularly affected by extreme observations, and it is important that the smallest group contains more cases than the number of predictor variables included in the model.

1. Multivariate normality: The predictor variables are assumed to follow a normal distribution within each category of the classification variable.

2. Equality of variance–covariance matrices (homogeneity): The variance and covariance structures of the predictors are expected to be comparable across all groups. This assumption can be evaluated using Box’s M test. In practice, linear discriminant analysis is generally preferred when covariance matrices are similar, whereas quadratic discriminant analysis is more appropriate when this condition is violated.
3. Independence: Observations are assumed to be obtained through random sampling, and the measurements from one individual are considered independent of those from any other individual.

2.7 Mahalanobis Distance

Mahalanobis distance is a scale-invariant measure that quantifies the separation between two multivariate distributions while accounting for correlations among variables. For two classes with mean vectors μ_1, μ_2 and a common covariance matrix Σ , the squared Mahalanobis distance is defined as:

$$D^2 = (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)^\top \boldsymbol{\Sigma}^{-1} (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2) .$$

This distance standardises the difference between the class means by the within-class variance–covariance structure. In classification, a larger Mahalanobis distance implies greater class separability and, consequently, higher expected predictive accuracy.

2.8 Relationship Between Logit and LDA

Both the logistic Regression and LDA model the posterior probabilities of the two population(response 0 and 1). The optimal Classifier rule is to classify observation $\mathbf{x}$ to either population(π_0 and π_1), i.e. to π_1 ,

$$P(G = 1|x) > P(G = 0|x) .$$

Using Bayes’ Theorem, this is equivalent to:

$$\frac{f_1(\mathbf{x})\pi_1}{f_0(\mathbf{x})\pi_0} > 1 .$$

Then taking log,

$$\log \frac{P(G = 1|\mathbf{x})}{P(G = 0|\mathbf{x})} = \log \frac{\pi_1}{\pi_0} + \log \frac{f_1(\mathbf{x})}{f_0(\mathbf{x})} > 0 ,$$

where π_k are the k th population's prior probabilities and $f_k(\mathbf{x})$ the class-conditional densities.

LDA assumes that the predictors follow a multivariate normal distribution within each class with a common covariance matrix:

$$(\mathbf{x} \mid G = k) \sim N(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}), \quad k = 0, 1.$$

Substituting the normal density into the log-odds and simplifying (using the symmetry of $\boldsymbol{\Sigma}^{-1}$) yields,

$$\log \frac{P(G = 1|\mathbf{x})}{P(G = 2|\mathbf{x})} = \underbrace{(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)^T \boldsymbol{\Sigma}^{-1}}_{\boldsymbol{\beta}^T} \mathbf{x} + \underbrace{-\frac{1}{2}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)^T \boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu}_1 + \boldsymbol{\mu}_2) + \log \frac{\pi_1}{\pi_2}}_{\beta_0}.$$

Thus, under the LDA assumptions, the log-odds is exactly a linear function of $\mathbf{x}$,

$$\log \frac{P(G = 1|\mathbf{x})}{P(G = 0|\mathbf{x})} = \beta_0 + \boldsymbol{\beta}^T \mathbf{x}.$$

Logistic regression, on the other hand, directly assumes that the log-odds is a linear function of the predictors. The parameters $(\beta_0, \boldsymbol{\beta})$ are estimated by maximizing the binomial likelihood without any distributional assumptions on $\mathbf{x}$. Hence, the two models share the same functional form for the log-odds. ([Johnson and Wichern, 2007](#); [James et al., 2021](#); [Hastie et al., 2009](#))

Chapter 3

Dataset and Variables

3.1 Data Source

The 2022 Bangladesh Demographic and Health Survey (2022 BDHS) is the ninth survey of its kind carried out in Bangladesh as part of the global DHS Program([National Institute of Population Research and Training \(NIPORT\) and ICF, 2023](#)). The survey was conducted under the supervision of the National Institute of Population Research and Training (NIPORT), Medical Education and Family Welfare Division, Ministry of Health and Family Welfare (MOHFW), Government of Bangladesh. The survey received funding from the Government of Bangladesh and the United States Agency for International Development (USAID).

The primary objective of the 2022 BDHS is to provide up-to-date estimates of basic demographic and health indicators for Bangladesh. Specifically, the survey collected comprehensive information on fertility and childhood mortality levels, fertility preferences, awareness and use of family planning methods, maternal and child health including breastfeeding practices, nutrition levels, and newborn care. The data collected by them are intended to assist national policymakers and various program managers in designing and evaluating strategies for improving the overall healthcare of the common people in Bangladesh. The survey also informs on indicators relevant to the Sustainable Development Goals (SDGs) for Bangladesh.

Sampling Design and Coverage

The sampling frame for the 2022 BDHS was the Integrated Multi-Purpose Sampling Master Sample, taken from a complete list of enumeration areas (EAs) covering the entire country. This master sample was prepared by the Bangladesh Bureau of Statistics (BBS) based on the 2011 population census. The frame contains information on EA location, residence type (city corporation, other urban, or rural), and estimated household counts. Bangladesh comprises eight administrative divisions: Barishal, Chattogram, Dhaka, Khulna, Mymensingh, Rajshahi, Rangpur, and Sylhet. The survey was designed to produce reliable estimates for key demographic and health indicators

separately for urban and rural areas within each division.

A two-stage stratified sampling design was employed. In the first stage, 675 EAs (237 urban, 438 rural) were selected with probability proportional to EA size. Following EA selection, a complete household listing operation in all selected EAs was conducted to create a sampling frame for the second stage. In the second stage, a systematic sample of 45 households was selected from each EA. This approach yielded a nationally representative sample. Ultimately, 30,018 households were successfully interviewed, with 30,078 women completing individual interviews (19,987 completing the full questionnaire). The survey achieved a 98 percent response rate and covered all de jure household members, all women aged 15-49, and all children aged 0-4.

3.2 Selected Variables

3.2.1 Dependent Variable

The dependent variable has been constructed for our study using other relevant variables.

Blood Pressure

sbp is a constructed variable representing the respondent's systolic blood pressure, derived from the raw male/female survey variables. **wbp24** contains Systolic blood pressure reading for women and **mbp24** contains the same reading for men.

Similarly **dbp** is a constructed variable representing the respondent's diastolic blood pressure using **wbp25** and **mbp25** for both genders.

Hypertension Diagnosis

The variable **wbp16** contains information whether respondent (woman), has ever been told by a health professional that she has hypertension. **mbp16** contains similar information about the male respondents.

ht_ever_diagnosed combines these two variables into a single variable for all respondents. Only non-missing values are included for this.

Response Variable

A binary outcome variable, **hypertension**, was constructed with a combination of information from variables blood pressure, blood glucose level and diagnosis.

We started with all respondents coded as 0 (not hypertensive).

Later, we marked the variable as 1 (hypertensive) if any of these conditions were found true:

1. If variable **ht_ever_diagnosed** = 1, it indicates that the respondent has been diagnosed with hypertension by a health professional.

2. A systolic blood pressure (**sbp**) greater than 140 mmHg is considered above the normal threshold and indicates elevated blood pressure.
3. A diastolic blood pressure (**dbp**) greater than 90 mmHg is considered above the normal threshold and indicates elevated blood pressure.

So, the variable captures information from both self-reported diagnosis and measured high blood pressure during the survey. It is noteworthy that the cleaned data only includes respondents with non-missing data.

3.2.2 Independent Variables

The six independent variables used in the study:

Wealth Index

The variable **hv270** composite measure of a household's cumulative living standard. It is used to categorize households into relative economic status groups. BDHS usually divides households into quintiles: Poorest, Poorer, Middle, Richer, Richest.

Wealth index is often used as a proxy for socioeconomic status in health and demographic studies, like analyzing inequalities in health outcomes.

Sex

The variable **hv104** is a categorical variable coded 1(male) and 2(female) which represents the biological sex of the respondent. It helps stratify outcomes by sex, e.g., prevalence of hypertension or diabetes among males vs females.

Age

hv105 variable is continuous which stores the age of the respondent in completed years at the time of the survey. Age is usually treated as a continuous variable for regression or can be categorized into age groups (e.g., 15–24, 25–34, 35–44, 45–49) for descriptive statistics. Age is critical for understanding health risks and demographic patterns.

Diabetes Diagnosis

The categorical variable **diab_ever_diagnosed** was constructed from the combination of two variables to account for both genders. 1 indicates respondent being diabetic. **sb236** referred us to whether the female respondents was ever diagnosed with diabetes. And variable **sb336** indicated whether the male respondents was ever diagnosed with diabetes.

Blood Glucose Level

The continuous variable `blood_glucose_mgdl` was constructed from the combination of two variables to account for both genders. `sb267` referred us to the Plasma blood glucose (mg/dL) measure of the female respondents. And variable `sb367` referred us to the same measure for male respondents.

Body Mass Index

Body Mass Index is a widely used measure to assess body fat based on weight relative to height. BMI is a simple indicator of whether a person is underweight, normal weight, overweight, or obese. It is commonly used in health surveys, like BDHS, to study risk factors for diseases such as hypertension, diabetes, and cardiovascular conditions. The continuous variable `bmi` is thus constructed from the combination of variables in the data `ha40` and `hb40` which contains the body mass index for both genders.

3.3 Selection Criteria

This study has heavily relied on past available literature for the selection of explanatory variables. The selection was primarily guided by previous empirical studies on hypertension and related cardiovascular risk factors. A recent study [Ahmed et al. \(2019\)](#) defined the response variable using the guidelines from the American Heart Association(AHA). Their methodology included Multi-level Logistic Regression analysis so that they could identify factors associated with undiagnosed hypertension. A study ([Al Kibria et al., 2021](#)) conducted prior to this study followed a similar track. Then another well known study([Chowdhury et al., 2016](#)) also defined the outcome variable and choose explanatory variables in a similar fashion. It is noteworthy that the authors in all these studies didn't choose a binary logistic regression analysis. The reason behind this is : BDHS does not use simple random sampling. It uses two-stage cluster sampling, enumeration Areas (EAs) or PSUs, households within clusters. Surveyed observations from the same cluster share similar environment, healthcare and socioeconomic conditions. Ignoring this clustering results in underestimated standard errors. Consequently giving smaller p-values, i.e. false significant variables. On the contrary, binary logistic regression does exactly that(ignores clustering) by assuming all observations are independent. But since our study does not involve observing the effects of clustering and estimating the determinants of hypertension using BDHS, we are free to carry out this experiment of fitting binary models to the said dataset.

Prior to model fitting, exploratory data analysis was conducted to examine the distribution of each variable and to assess potential multicollinearity. Highly correlated predictors were carefully evaluated to avoid redundancy and to satisfy the assumptions of Linear Discriminant Analysis (LDA). Variable selection was also constrained by data availability within the BDHS dataset.

Chapter 4

Findings

4.1 Simulation Result

To evaluate the comparative performance of logistic regression and linear discriminant analysis under controlled conditions, we conducted a Monte Carlo simulation study. Data were generated from a logistic model with four predictors (Treatment, Age, Sex, Baseline SBP) with known coefficients. The true probabilities of the outcome were used to compute the true values of three performance indices: the C-index (concordance probability), the Q-index (a logarithmic scoring rule), and the B-index (related to the Brier score). For each simulated dataset, the sample was randomly split into a training set (70 %) and a test set (30%). Models were fitted on the training data and predictions were evaluated on the test set. The simulation was repeated 1000 times for each scenario. Three factors were manipulated: sample size n , correlation ρ between the continuous predictors Age and SBP, and the Mahalanobis distance between the two classes.

We fixed the correlation at $\rho = 0.5$ and varied the sample size: $n = 40, 60, 100, 200, 1000$. Tables 4.1–4.3 present the mean C, Q and B indices for logistic regression and LDA, together with the true value of each index (based on the known data-generating probabilities).

4.1.1 Performance Under Different Sample Size

C-index Comparison(different n)

Table 4.1: C-index comparison (different n)

SampleSize	LOGIT_C_mean	LDA_C_mean
40	0.627	0.626
60	0.639	0.639
100	0.661	0.661
200	0.672	0.671
1000	0.684	0.684

Across all sample sizes, both Logistic Regression and LDA show increasing performance as the sample size grows. The true C-index is 0.68 (n=1000). Logistic and LDA are nearly identical in all scenarios.

Q-index Comparison(different n)

Table 4.2: Q-index comparison (different n)

SampleSize	LOGIT_Q_mean	LDA_Q_mean
40	-0.236	-0.117
60	-0.047	-0.032
100	0.028	0.031
200	0.057	0.057
1000	0.081	0.081

The true value of Q index is 0.0859(n=1000). Logistic shows slightly lower Q-index at small sample sizes but converges with LDA at larger n.

B-index Comparison(different n)

Table 4.3: B-index Comparison(different n)

SampleSize	LOGIT_B_mean	LDA_B_mean
40	0.731	0.735
60	0.747	0.749
100	0.763	0.763
200	0.770	0.770
1000	0.776	0.776

Again, both models perform similarly in case of B index, with marginal differences; LDA slightly outperforms Logistic at smaller n. True B-index is found to be 0.778(n=1000).

Proportion of Performance(different n)

Across the simulated sample sizes (n = 40, 60, 100, 200, 1000), Logistic regression outperforms LDA in less than 52% of replications, even at the largest sample size. As the sample size increases, the proportion of replications where Logistic performs better gradually rises (up to 51% for n = 1000), but still does not reach a clear majority.

For smaller samples (n = 40, 60), Logistic regression only outperforms LDA in 19–25% of replications for the C-index, indicating that with limited data, LDA is often equally or more

reliable. while for the Q and B indices, the proportion is slightly higher, ranging from 35–52% as sample size increases. This suggests that even for Q and B, Logistic regression does not consistently dominate LDA, but again the differences diminish with larger samples.

SampleSize	Prop_Logit_Better_C	Prop_Logit_Better_Q	Prop_Logit_Better_B
40	0.194	0.350	0.383
60	0.252	0.393	0.409
100	0.321	0.453	0.461
200	0.419	0.507	0.527
1000	0.514	0.516	0.521

Table 4.4: Proportion of replications where Logistic regression outperforms LDA(different n)

4.1.2 Performance Under Different Levels of Predictor Correlation

C-index Comparison(different ρ)

Table 4.5: C-index Comparison(different ρ)

rho	True_C_mean	LOGIT_C_mean	LDA_C_mean
0.3	0.679	0.675	0.675
0.6	0.693	0.689	0.689
0.9	0.705	0.702	0.702

Predictive accuracy increases slightly as the correlation between predictors (ρ) rises. Logistic regression and LDA perform very similarly, with differences in mean C-index negligible.

Q-index Comparison(different ρ)

Table 4.6: Q-index Comparison(different ρ)

rho	True_Q_mean	LOGIT_Q_mean	LDA_Q_mean
0.3	0.077	0.072	0.072
0.6	0.090	0.084	0.084
0.9	0.101	0.096	0.096

Higher correlations also lead to slightly better Q-index values. Again, both models show comparable performance, with Logistic regression marginally lower than LDA at lower values of ρ and marginally higher than LDA at higher values of ρ .

B-index Comparison(different ρ)

Table 4.7: B-index Comparison(different ρ)

rho	True_B_mean	LOGIT_B_mean	LDA_B_mean
0.3	0.775	0.774	0.774
0.6	0.779	0.777	0.777
0.9	0.783	0.781	0.781

The B-index increases with increasing predictor correlation, indicating improved overall fit. Differences between Logistic regression and LDA remain minimal, suggesting both models are robust to changes in predictor correlation.

Proportion of Performance(different ρ)

Across the simulated correlation levels ($\rho = 0.3, 0.6, 0.9$), Logistic regression slightly outperforms LDA in about 51–52% of replications for the C-index, indicating that the two models are generally comparable regardless of predictor correlation. For the Q- and B-indices, Logistic regression shows a slightly higher proportion of better performance at lower correlations ($\rho = 0.3$ – 0.6 , 51–56%), but this advantage diminishes at the highest correlation ($\rho = 0.9$), where it barely exceeds 51% for both indices.

Overall, these results suggest that predictor correlation has only a minor effect on the relative performance of Logistic regression versus LDA, and neither model consistently dominates the other across different correlation levels.

Table 4.8: Proportion of replications where Logistic regression outperforms LDA (different ρ)

rho	Prop_Logit_Better_C	Prop_Logit_Better_Q	Prop_Logit_Better_B
0.3	0.516	0.553	0.565
0.6	0.516	0.548	0.556
0.9	0.522	0.513	0.514

4.1.3 Performance Under Different Mahalanobis Distance

C-index Comparison(different D^2)

Table 4.9: C-index Comparison(different D^2)

maha	True_C_mean	LOGIT_C_mean	LDA_C_mean
0.5	0.638	0.633	0.633
1.0	0.759	0.757	0.757
1.5	0.854	0.852	0.852
2.0	0.921	0.920	0.920

Both Logit and LDA closely follow the true C-index values across all distances. As the distance increases (groups are more separated), the C-index rises, reflecting easier discrimination. The difference between Logit and LDA is minimal (less than 0.001), showing that both models perform almost identically in terms of C-index.

Q-index Comparison(different D^2)

Table 4.10: Q-index Comparison(different D^2)

maha	True_Q_mean	LOGIT_Q_mean	LDA_Q_mean
0.5	0.043	0.037	0.037
1.0	0.159	0.154	0.154
1.5	0.316	0.310	0.310
2.0	0.485	0.479	0.480

Logit and LDA again are very close to the true Q values, slightly underestimating them. With increasing distance, the Q-index increases, indicating that discrimination improves as the groups become more separated. It is noticeable that the differences between Logit and LDA remain negligible.

B-index Comparison (different D^2)

Table 4.11: B-index Comparison(different D^2)

maha	True_B_mean	LOGIT_B_mean	LDA_B_mean
0.5	0.764	0.762	0.762
1.0	0.801	0.799	0.799
1.5	0.844	0.843	0.843
2.0	0.887	0.886	0.886

Increasing distance improves the B-index for both models, consistent with better separability of groups. Again, the performance between Logit and LDA is almost identical.

Proportion of Performance (different D^2)

Across the simulated group separation levels (distance = 0.5, 1.0, 1.5, 2.0), LDA slightly outperforms Logit in roughly 50% of replications for the C-index when distance is low (0.5), indicating that the two models are generally comparable at minimal group separation. As distance increases, the better performance of logit decreases to 42–48%, suggesting that LDA may perform slightly better at higher separations given normality. For the Q- and B-indices, Logistic regression shows a small advantage at lower distance (0.5–1.0, 46–50%), but this advantage diminishes as distance

increases, dropping to 39–38% at distance = 2.0.

Alltogether,again, these results indicate that group separation has only a modest effect on the relative performance of Logistic regression compared to LDA, and LDA mildly dominates the simulated scenario. Logistic regression appears to be under performing at higher separations because the two additional predictor variables included in the model do not contribute to the simulated group separation, ultimately reducing its relative performance compared with LDA.

Table 4.12: Proportion of simulations where Logistic Regression outperforms LDA(different D^2)

maha	Prop_Logit_Better_C	Prop_Logit_Better_Q	Prop_Logit_Better_B
0.5	0.495	0.495	0.502
1.0	0.468	0.466	0.474
1.5	0.483	0.449	0.461
2.0	0.427	0.391	0.386

4.2 Empirical Validation: Comparative Evaluation Using BDHS Data

4.2.1 Exploratory Analysis

We are to assess the independent variables to get an idea about their distribution.

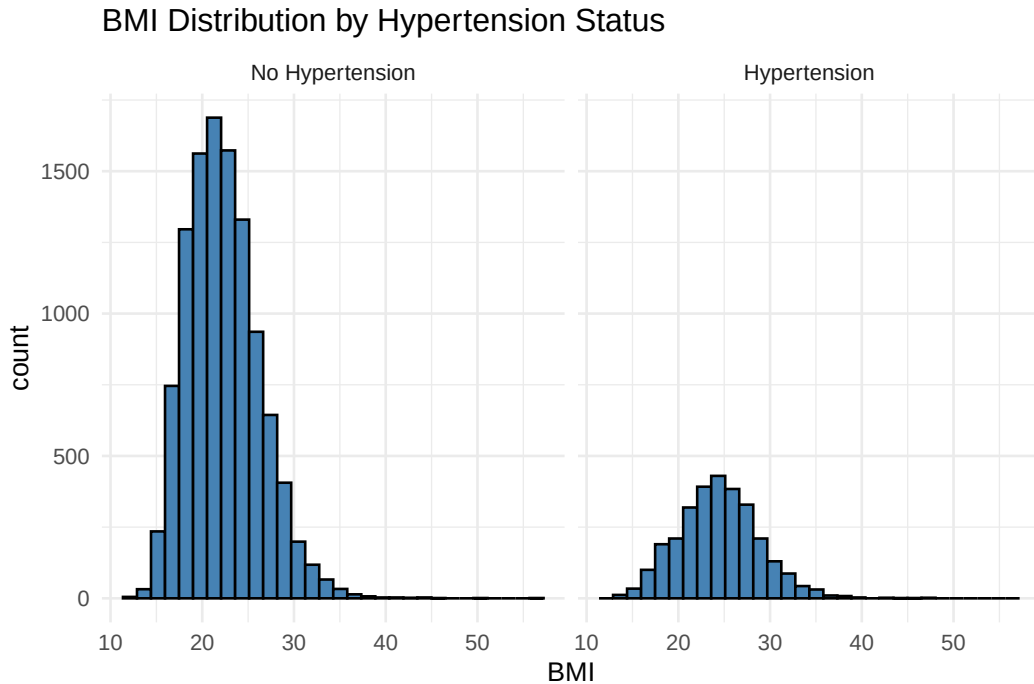


Figure 4.1: Histogram for Body Mass Index

The figure 4.1 helps to identify the normality of body mass index for both classes. But it

is important to remember that LDA is susceptible to outliers which is evident in the variable. Outliers can shift the decision boundary incorrectly. Again, they can increase variance estimates, making the estimated covariance larger. This can make classes appear more overlapping than they actually are. So classification accuracy drops.

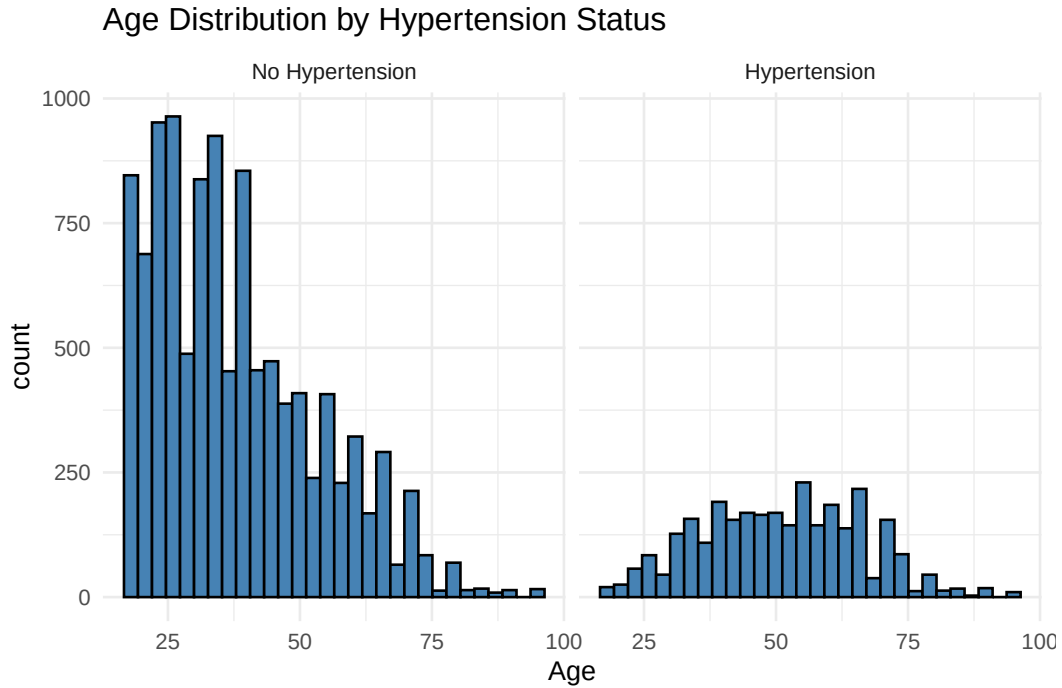


Figure 4.2: Histogram for Age

The independent variable in figure 4.2 representing age of the population is highly skewed. This can pose problems for LDA. Because of this, the class conditional distribution are not normal which means distortion of estimated mean vectors of each class(0 and 1) and the estimated shared covariance matrix. Consequently, the linear decision boundary may not be optimal and LDA may misclassify points, especially in the tail region. Thus, the predictive capability should decrease.

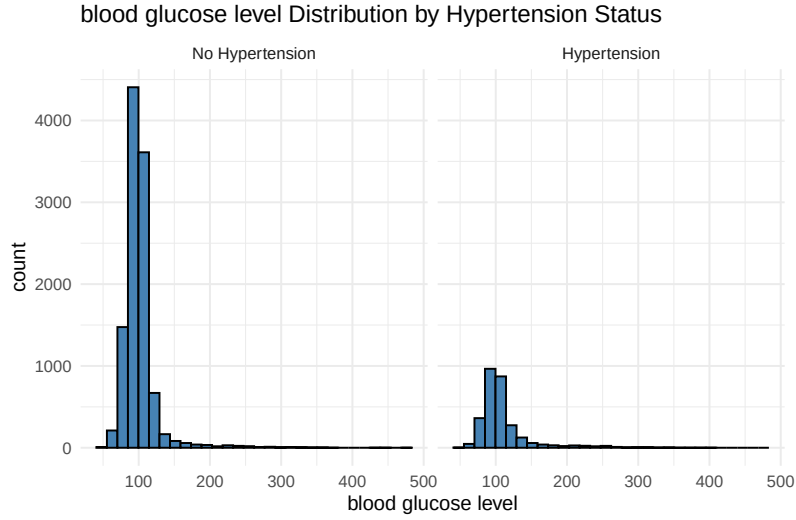


Figure 4.3: Histogram for Blood Glucose level

The outlier problems is also visible in this variable (figure 4.3). Otherwise it seems normally distributed.

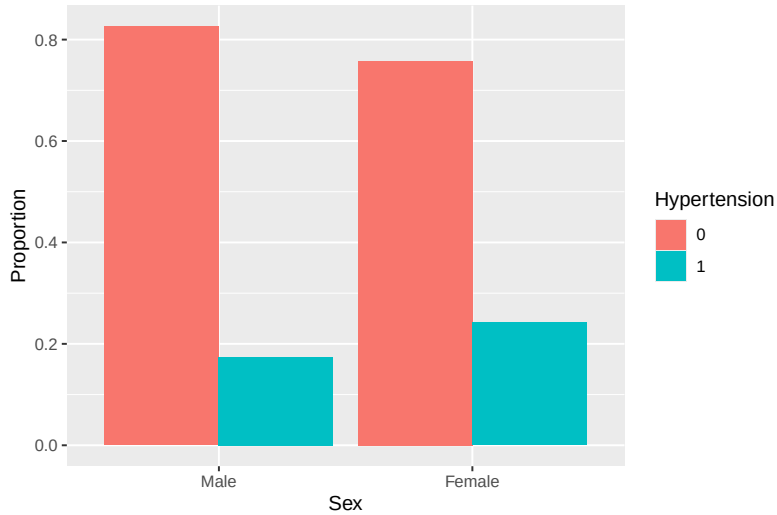


Figure 4.4: Bar-plot for Sex

The purpose of a bar-plot in figure 4.4 was to observe the variation for genders. The proportion of male belonging to a class say hypertensive or not is equivalent to the proportion of female of the same class. But it has led us towards another difficulty for LDA which is class imbalance. This is very common in BDHS data. From all the figure 4.1-4.4, it is evident that for the dependent variable, one class (non-hypertensive) dominates over the other (hypertensive). LDA computes class priors, often from the data:

$$\hat{\pi}_k = \frac{n_k}{n} ,$$

where n_k is the number of observations in class k .

The decision boundary is influenced by the priors; classes with more observations pull the

boundary toward the minority class. LDA may classify almost everything as class(non hypertensive i.e.0) to maximize overall accuracy and the minority class (hypertensive i.e. 1) is often underpredicted. As a result we see a poor sensitivity for minority class. A similar scenario can be seen for Wealth Index.



Figure 4.5: Bar-plot for Wealth Index

4.2.2 Fitted Models

Logistic Regression and LDA models were fitted to the BDHS dataset by gradually increasing the number of explanatory variables in a way to suit our comparison study as shown in Figure 4.5 through their ROC curves. Both models have shown good discriminative capacity along the process as they offer similar performance in their prediction capability as we keep on including significant variables. Note that we have not handled extreme values, skewed variables and unequal response of our data. Even with such delicacies, the discriminatory capability remains almost similar.

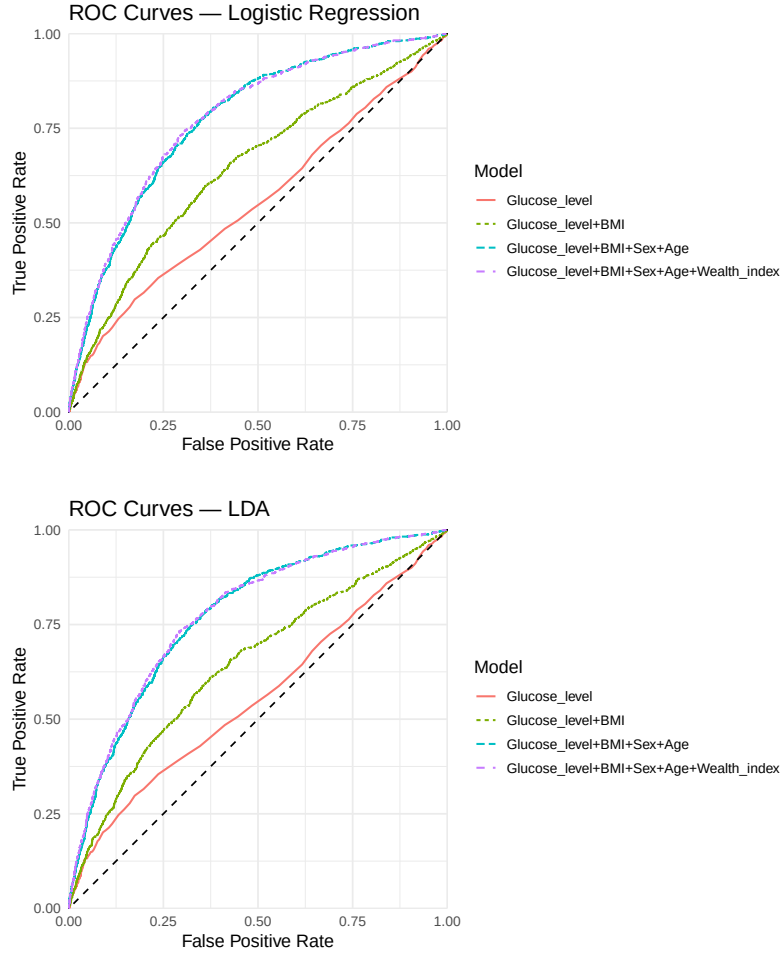


Figure 4.6: ROC Curves for various fits

Since the ROC curves for logistic regression and LDA (Figure 4.5) are virtually indistinguishable across all predictor combinations, we can say that the choice of one of these model won't have consequential difference in classification. The near-perfect overlap of the LDA and Logistic densities reinforces the idea that both methods are giving essentially the same predictions for each individual.

Figure 4.6, the two plots show distribution of the linear predictor scores from the LDA and Logistic models for the full set of predictor, separately for observations with hypertension and without. The figures are largely similar, indicating that both models assign similar risk scores for specific group(0 and 1). When we have two class distributions overlap like this(for each model), perfect separation is mathematically impossible. Unlike our artificially created datasets, real health data does not form clean clusters. So the overlap was always expected. But what is noteworthy is that both models are extracting similar linear separation from the data. So, even if the data is roughly linearly separable (but weakly), both models will produce similar scores.

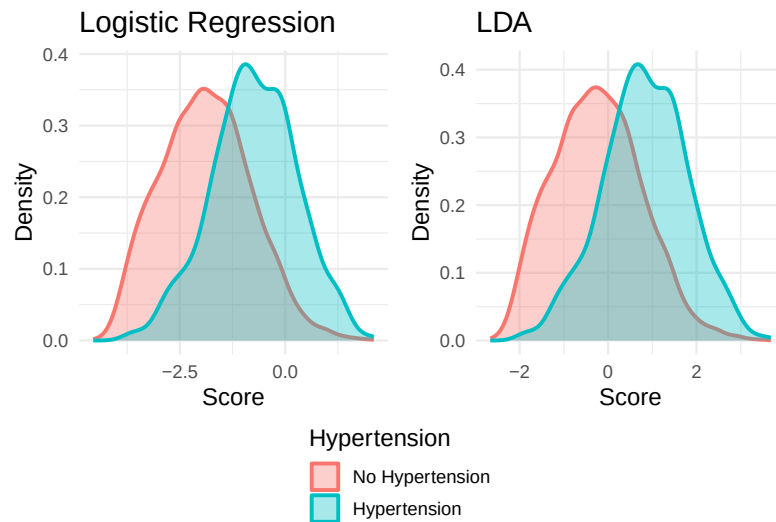


Figure 4.7: Score Comparison

4.2.3 Bootstrapping

Bootstrap is non-parametric technique and works well even when theoretical distributions are complicated. It can quantify the reliability of our performance estimates and checks whether one model consistently outperforms the other. It can help estimate confidence intervals for the AUC of each model and for their difference, giving a range of plausible values. So we can visualize how the AUC varies across resamples.

The bootstrap results provide additional evidence that there is no meaningful difference in predictive performance between logistic regression and LDA. This closes the deal and helps us reach our conclusion.

Table 4.13: Bootstrap 95% Confidence Intervals for AUC (Full Model)

Model	AUC	2.5%	97.5%
Logistic Regression	0.7851	0.7682	0.8012
LDA	0.7855	0.7684	0.80197
Difference	-0.0004	-0.0018	0.0009

1. Overlapping confidence intervals: The 95% confidence intervals for the AUC of Logistic (0.77–0.8) and LDA (0.77–0.8) overlap completely. This means that the true AUC of either model could lie anywhere in that range, and the intervals are nearly identical.
2. Difference includes zero: The estimated difference (Logistic AUC – LDA AUC) is only 0.0004, and its 95% confidence interval contains zero. Therefore, we cannot conclude that one model outperforms the other; the observed tiny difference is well within sampling variability.

Chapter 5

Conclusion

5.1 Discussion

This study set out to investigate the comparative predictive performance of two longstanding workhorses of statistical classification: Logistic Regression and Linear Discriminant Analysis. Through a structured Monte Carlo simulation and an empirical validation using the 2022 Bangladesh Demographic and Health Survey, we aimed to provide clarity on whether one method consistently outperforms the other, or if their performance is contingent upon specific data characteristics.

The findings from our simulation study were remarkably consistent. Across a wide spectrum of controlled conditions—varying sample sizes from as few as 40 to as many as 1000 observations, altering the correlation between predictors from low (0.3) to very high (0.9), and manipulating the Mahalanobis distance to represent different levels of class separability; both Logistic Regression and LDA exhibited nearly identical predictive accuracy. The C-index, Q-index, and B-index values for the two models were, for all practical purposes, indistinguishable. The proportions of replications where one model outperformed the other rarely deviated far from 50%, reinforcing the conclusion that neither method holds a universal advantage under the normally distributed data conditions we simulated. This equality was most evident in the C-index, where the mean values for Logit and LDA were often within a margin of 0.001.

This finding of near-equivalence was strongly corroborated by our empirical analysis using the BDHS dataset. When applied to a real-world public health problem—predicting hypertension—the ROC curves for Logistic Regression and LDA were virtually superimposed, regardless of the number of predictors included in the model. Both models assign similar risk scores to individuals as well. Most definitively, the bootstrap analysis, a robust non-parametric technique, confirmed these visual impressions. The 95% confidence intervals for the Area Under the Curve (AUC) for both models overlapped almost completely, and the confidence interval for their differ-

ence comfortably contained zero. This provides strong statistical evidence that, for this dataset and set of predictors, the models are essentially interchangeable in terms of their discriminative ability.

Our work is deeply rooted in the foundational and contemporary research that has explored this comparison over the decades. The theoretical groundwork laid by [Efron \(1975\)](#), which highlighted the asymptotic efficiency of both methods and the robustness of logistic regression, provided the initial impetus for our study. Our simulation results align perfectly with the conclusions of [Pohar et al. \(2004\)](#) and [Lei and Koehly \(2003\)](#), who also found that when the assumptions of LDA (particularly multivariate normality) are reasonably satisfied, its performance is on par with logistic regression. Furthermore, our empirical findings echo those of [Panagiotakos \(2006\)](#), who, using a real-world epidemiological dataset, found no statistical difference between the ROC curves of the two models, a result we were able to confirm and strengthen through our bootstrap analysis. By systematically varying key data parameters in our simulation and validating those results with a complex, real-world survey dataset, our study not only confirms the existing body of knowledge but also extends it by providing a more granular, scenario-specific confirmation of this long-standing statistical equivalence. We acknowledge and thank these authors for their invaluable contributions that have shaped the direction and methodology of this research.

5.2 Future Research Direction

While this study provides a robust answer to the question of comparative performance under common conditions, it also opens several avenues for future investigation.

First, the simulation framework could be expanded to challenge the models further. Future research should explore scenarios where LDA’s core assumptions are systematically and severely violated. This includes generating data from non-normal distributions (e.g., skewed or heavy-tailed distributions) and simulating groups with unequal covariance matrices. Under such conditions, the robustness of logistic regression might become more pronounced, and the performance gap between the two methods could widen. Investigating the impact of outliers and highly multicollinear predictors would also provide valuable insights for practitioners dealing with messy real-world data.

Second, and perhaps most excitingly, is the potential to refine the simulation methodology itself. A key limitation of our simulation, and indeed of most traditional Monte Carlo studies, is that the data is generated from a pre-specified, often simplistic, statistical model. To create synthetic data that more faithfully captures the complexity and nuances of real-world populations, future studies could leverage advanced generative models, such as **Variational Autoencoders (VAEs)**. A VAE could be trained on a real dataset like the BDHS to learn the underlying probability distribution of the data, including all its complex correlations, non-linearities, and noise

structures. Once trained, the VAE can generate an infinite number of new, synthetic datasets that retain the statistical essence of the original. These synthetic datasets would be far more realistic than those generated by a simple logistic model. By applying both Logistic Regression and LDA to these VAE-generated datasets, researchers could compare their performance in an environment that is both controlled (since we can generate as much data as needed) and realistic (since the data mimics the complexity of the real world). This approach would provide a much more stringent and meaningful test of model performance, bridging the gap between theoretical simulations and the unpredictability of empirical data. This represents a promising frontier for comparative research in statistics and machine learning.

5.3 Limitations

While this study provides a comprehensive comparison of Logistic Regression and Linear Discriminant Analysis under various controlled conditions, several limitations should be acknowledged when interpreting the findings and considering their generalizability to real-world applications.

First, the Monte Carlo simulation, while powerful, is inherently constrained by the data-generating process we chose. Our primary simulation framework assumed a logistic model as the true underlying process. This setup naturally favors logistic regression when its assumptions are met, yet we observed comparable performance from LDA. However, this framework cannot capture the infinite complexity of real-world data. We can never be entirely certain about the true multivariate distribution of any empirical dataset. Consequently, the simulation results provide a benchmark under idealized conditions but may not fully reflect performance in every practical scenario.

Second, the simulation scenarios, while systematically varied, are not exhaustive. We manipulated sample size, predictor correlation, and Mahalanobis distance, but we did not explore all possible assumption violations. For instance, we did not simulate data with extreme multicollinearity, interaction effects between predictors, or more complex non-linear relationships. Furthermore, the predictors in our simulation were generated from multivariate normal distributions, which aligns with LDA’s assumptions. A more challenging and perhaps more realistic comparison would involve generating data from non-normal distributions (e.g., skewed or heavy-tailed distributions) or with unequal covariance matrices between groups—conditions under which LDA is known to be less robust. Future studies could expand the simulation framework to systematically violate LDA’s assumptions more severely.

Third, the variable selection process for the empirical validation using the BDHS dataset was guided by existing literature and subject-matter knowledge. While this is a standard and valid approach, it does not guarantee that the chosen set of predictors is optimal for either classification

method. A different set of variables, or the inclusion of interaction terms and non-linear transformations, might have yielded a different performance dynamic between the two models. The analysis did not employ automated variable selection techniques, which could have introduced another layer of complexity and potential bias.

Fourth, the quality and nature of the BDHS data itself impose limitations. As with most large-scale surveys, the data are subject to potential reporting biases (e.g., self-reported diabetes diagnosis), measurement errors in blood pressure and glucose levels, and missing data, despite our efforts to clean the dataset. The cross-sectional design of the BDHS also means we can only assess association, not prediction over time. The constructed hypertension outcome, while carefully defined, is a simplification of a complex clinical condition.

Finally, our study focused on a comparison of only two specific classifiers. The machine learning landscape offers a vast array of other powerful algorithms, such as random forests, support vector machines, and gradient boosting methods, which often exhibit high predictive accuracy and flexibility. The conclusion that LDA and logistic regression perform similarly does not imply that they are the best possible models for this classification task. It merely suggests that, given the data and the scenarios considered, the choice between these two specific, traditionally-used methods is not critical for predictive performance.

In summary, while this study offers valuable insights, its findings are bounded by the assumptions of its simulation design, the limitations of the real-world dataset, and the narrow focus on two specific models. The ultimate challenge remains: for any given real-world dataset, one may still need robust methods, potentially including more complex generative models or resampling techniques, to confidently determine which analytical approach provides the best discrimination.

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Appendix

R-codes

```
sample_sizes<-c(40,60,100,200,1000) #5 scenarios
rho<-c(.3,.6,.9)#3scenarios
n_replications<-1000
```

A function to calculate C index

```
Index_C <- function(P, Y) {
  concordant_model <- 0

  for (i in which(Y == 0)) {
    for (j in which(Y == 1)) {
      if (P[j] > P[i]) {
        concordant_model <- concordant_model + 1
      } else if (P[i] == P[j]) {
        concordant_model <- concordant_model + 0.5
      }
    }
  }

  return(concordant_model)
}
```

A function to run simulation

```
run_simulation <- function(n, rho, train_prop) {

  set.seed(NULL)

  # DATA
  Sigma <- matrix(
    c(
      10^2, rho * 10 * 15,
      rho * 10 * 15, 15^2
    ),
    nrow = 2
  )

  correlated_vars <- mvrnorm(n, mu = c(55, 130), Sigma = Sigma)

  basic_clinical_data <- data.frame(
    PatientID = 1:n,
    Treatment = sample(c("Placebo", "Drug"), n, replace = TRUE),

```

```

    Age = round(correlated_vars[, 1]),
    Sex = sample(c("Male", "Female"), n, replace = TRUE, prob = c(0.55, 0.45)),
    Baseline_SBP = round(correlated_vars[, 2])
  )

  beta_intercept <- -5.5
  beta_treatment <- 0.29
  beta_age <- 0.05
  beta_sex <- 0.37
  beta_sbp <- 0.02

  basic_clinical_data$log_odds <- beta_intercept +
    beta_treatment * (basic_clinical_data$Treatment == "Drug") +
    beta_age * basic_clinical_data$Age +
    beta_sex * (basic_clinical_data$Sex == "Female") +
    beta_sbp * basic_clinical_data$Baseline_SBP

  basic_clinical_data$probability <- 1 / (1 + exp(-basic_clinical_data$log_odds))

  basic_clinical_data$Response <- rbinom(n,
                                          size = 1,
                                          prob = basic_clinical_data$probability)

  # SPLIT into training and testing
  train_idx <- sample(1:n, size = round(n * train_prop))
  train_data <- basic_clinical_data[train_idx, ]
  test_data <- basic_clinical_data[-train_idx, ]

  # empty results
  results <- list(
    C_true = NA, Q_true = NA, B_true = NA,
    C_lda = NA, Q_lda = NA, B_lda = NA,
    C_logit = NA, Q_logit = NA, B_logit = NA
  )

  # TRUE VALUE
  P_true <- test_data$probability
  Y <- test_data$Response
  n_1 <- sum(Y == 1)
  n_0 <- sum(Y == 0)

  concordant_true <- Index_C(P_true, Y)
  results$C_true <- concordant_true / (n_1 * n_0)
  results$Q_true <- 1 + mean(Y * log2(P_true) + (1 - Y) * log2(1 - P_true))
  results$B_true <- 1 - (sum((P_true - Y)^2) / length(Y))

```

```

# LDA
tryCatch({
  lda_model <- lda(Response ~ Treatment + Age + Sex + Baseline_SBP,
                  data = train_data)
  lda_predictions <- predict(lda_model, test_data)
  P_lda <- lda_predictions$posterior[, 2]

  concordant_lda <- Index_C(P_lda, Y)
  results$C_lda <- concordant_lda / (n_1 * n_0)
  results$Q_lda <- 1 + mean(Y * log2(P_lda) + (1 - Y) * log2(1 - P_lda))
  results$B_lda <- 1 - (sum((P_lda - Y)^2) / length(Y))
})

# Logistic
tryCatch({
  logit_model <- glm(
    Response ~ Treatment + Age + Sex + Baseline_SBP,
    data = train_data,
    family = binomial
  )

  P_logit <- predict(logit_model, newdata = test_data, type = "response")
  concordant_logit <- Index_C(P_logit, Y)

  results$C_logit <- concordant_logit / (n_1 * n_0)
  results$Q_logit <- 1 + mean(Y * log2(P_logit) + (1 - Y) * log2(1 - P_logit))
  results$B_logit <- 1 - (sum((P_logit - Y)^2) / length(Y))
})

return(
  data.frame(
    SampleSize = n,
    rho = rho,
    true_C = results$C_true,
    true_Q = results$Q_true,
    true_B = results$B_true,
    LDA_C = results$C_lda,
    LDA_Q = results$Q_lda,
    LDA_B = results$B_lda,
    LOGIT_C = results$C_logit,
    LOGIT_Q = results$Q_logit,
    LOGIT_B = results$B_logit
  )
)
}

```

1000 times data generated each for different sample size $n = 40, 60, 100, 200$

```
# Empty storage
all_results <- data.frame()

for (n in sample_sizes) {
  for (rep in 1:n_replications) {

    result <- run_simulation(n, 0.5, 0.7)
    result$Replication <- rep

    # Append to the main results data frame
    all_results <- rbind(all_results, result)

  }
}
```

for each sample size $i=40,60,100,200$ we find the mean index and then find which model perform better

```
# Comparison of index averages
Comparison_of_indexes <- all_results %>%
  group_by(SampleSize) %>%
  summarise(
    # C averages
    True_C_mean = mean(true_C, na.rm = TRUE),
    LDA_C_mean = mean(LDA_C, na.rm = TRUE),
    LOGIT_C_mean = mean(LOGIT_C, na.rm = TRUE),

    # Q averages
    True_Q_mean = mean(true_Q, na.rm = TRUE),
    LDA_Q_mean = mean(LDA_Q, na.rm = TRUE),
    LOGIT_Q_mean = mean(LOGIT_Q, na.rm = TRUE),

    # B averages
    True_B_mean = mean(true_B, na.rm = TRUE),
    LDA_B_mean = mean(LDA_B, na.rm = TRUE),
    LOGIT_B_mean = mean(LOGIT_B, na.rm = TRUE)
  )

# Proportion of replications where Logistic regression performs better than LDA
Better_performance_of_logit <- all_results %>%
  group_by(SampleSize) %>%
  summarise(
    Prop_Logit_Better_C = mean(LOGIT_C > LDA_C, na.rm = TRUE),
    Prop_Logit_Better_Q = mean(LOGIT_Q > LDA_Q, na.rm = TRUE),
  )
```

```

    Prop_Logit_Better_B = mean(LOGIT_B > LDA_B, na.rm = TRUE)
  )

```

TABLE

```

last_true_C <- Comparison_of_indexes$True_C_mean[which.max(Comparison_of_indexes
                                                         $SampleSize)]
last_true_Q <- Comparison_of_indexes$True_Q_mean[which.max(Comparison_of_indexes
                                                         $SampleSize)]
last_true_B <- Comparison_of_indexes$True_B_mean[which.max(Comparison_of_indexes
                                                         $SampleSize)]

```

```

Comparison_of_indexes %>%
  select(SampleSize, LOGIT_C_mean, LDA_C_mean) %>%
  gt() %>%
  tab_header(
    title = "C-index Comparison",
    subtitle = paste("True C-index is", round(last_true_C, 3))
  ) %>%
  fmt_number(
    columns = -SampleSize,
    decimals = 3
  )

```

```

Comparison_of_indexes %>%
  select(SampleSize, LOGIT_Q_mean, LDA_Q_mean) %>%
  gt() %>%
  tab_header(
    title = "Q-index Comparison",
    subtitle = paste("True Q-index is", round(last_true_Q, 3))
  ) %>%
  fmt_number(
    columns = -SampleSize,
    decimals = 3
  )

```

```

Comparison_of_indexes %>%
  select(SampleSize, LOGIT_B_mean, LDA_B_mean) %>%
  gt() %>%
  tab_header(
    title = "B-index Comparison",
    subtitle = paste("True B-index is", round(last_true_B, 3))
  ) %>%
  fmt_number(
    columns = -SampleSize,
    decimals = 3
  )

```

```

)

Better_performance_of_logit %>%
  gt() %>%
  tab_header(
    title = "Proportion of Replications Where Logistic Performs Better"
  ) %>%
  fmt_number(
    columns = -SampleSize,
    decimals = 3
  )

```

1000 times data generated each for different sample correlation $\rho = .3, .6, .9$

```

# Empty storage
all_results <- data.frame()

for (rho_value in rho) {
  for (rep in 1:n_replications) {

    result <- run_simulation(1000, rho_value, 0.7)
    result$Replication <- rep

    # Append the result to the main data frame
    all_results <- rbind(all_results, result)

  }
}

```

for each correlation $\rho = .3, .6, .9$ we find the mean index and then find which model perform better

```

# Comparison of index averages by rho
Comparison_of_indexes <- all_results %>%
  group_by(rho) %>%
  summarise(
    # C averages
    True_C_mean = mean(true_C, na.rm = TRUE),
    LDA_C_mean = mean(LDA_C, na.rm = TRUE),
    LOGIT_C_mean = mean(LOGIT_C, na.rm = TRUE),

    # Q averages
    True_Q_mean = mean(true_Q, na.rm = TRUE),
    LDA_Q_mean = mean(LDA_Q, na.rm = TRUE),
    LOGIT_Q_mean = mean(LOGIT_Q, na.rm = TRUE),
  )

```

```

    # B averages
    True_B_mean    = mean(true_B, na.rm = TRUE),
    LDA_B_mean     = mean(LDA_B, na.rm = TRUE),
    LOGIT_B_mean   = mean(LOGIT_B, na.rm = TRUE)
  )

  # Proportion of replications where Logistic regression performs better than LDA
  Better_performance_of_logit <- all_results %>%
    group_by(rho) %>%
    summarise(
      Prop_Logit_Better_C = mean(LOGIT_C > LDA_C, na.rm = TRUE),
      Prop_Logit_Better_Q = mean(LOGIT_Q > LDA_Q, na.rm = TRUE),
      Prop_Logit_Better_B = mean(LOGIT_B > LDA_B, na.rm = TRUE)
    )

```

TABLE

```

Comparison_of_indexes %>%
  select(rho, True_C_mean, LOGIT_C_mean, LDA_C_mean) %>%
  gt() %>%
  tab_header(title = "C-index Comparison by Correlation") %>%
  fmt_number(columns = -rho, decimals = 3)

Comparison_of_indexes %>%
  select(rho, True_Q_mean, LOGIT_Q_mean, LDA_Q_mean) %>%
  gt() %>%
  tab_header(title = "Q-index Comparison by Correlation") %>%
  fmt_number(columns = -rho, decimals = 3)

Comparison_of_indexes %>%
  select(rho, True_B_mean, LOGIT_B_mean, LDA_B_mean) %>%
  gt() %>%
  tab_header(title = "B-index Comparison by Correlation") %>%
  fmt_number(columns = -rho, decimals = 3)

Better_performance_of_logit %>%
  gt() %>%
  tab_header(
    title = "Proportion Logistic Better (by Correlation)"
  ) %>%
  fmt_number(columns = -rho, decimals = 3)

```

Maha distance

```
maha_values <- c(0.5, 1, 1.5, 2) # 4 scenarios
```

```
# Function to compute scaling factor for a given direction
```

```
get_mean_shift <- function(D, direction, Sigma) {  
  v <- direction / sqrt(sum(direction^2)) # unit vector  
  k <- D / sqrt(t(v) %*% solve(Sigma) %*% v)  
  return(k * v)  
}
```

```
# Modified simulation function
```

```
run_simulation_maha <- function(n, rho, maha_dist, train_prop = 0.7) {  
  # Covariance matrix (same as before)  
  Sigma <- matrix(c(10^2, rho*10*15, rho*10*15, 15^2), nrow = 2)  
  
  # Choose a direction (e.g., both variables increase together)  
  direction <- c(1, 1)  
  Delta <- get_mean_shift(maha_dist, direction, Sigma)  
  
  mu0 <- c(55, 130) # group 0 mean  
  mu1 <- mu0 + Delta # group 1 mean  
  
  # Generate group labels (balanced for simplicity)  
  n_per_group <- round(n / 2)  
  Y <- c(rep(0, n_per_group), rep(1, n - n_per_group))  
  
  # Generate predictors conditionally  
  X0 <- MASS::mvrnorm(n_per_group, mu = mu0, Sigma = Sigma)  
  X1 <- MASS::mvrnorm(n - n_per_group, mu = mu1, Sigma = Sigma)  
  X <- rbind(X0, X1)  
  
  # Create data frame (add any other covariates if desired)  
  data <- data.frame(  
    Response = Y,  
    Age = round(X[, 1]),  
    Baseline_SBP = round(X[, 2]),  
    Treatment = sample(c("Placebo", "Drug"), n, replace = TRUE),  
    Sex = sample(c("Male", "Female"), n, replace = TRUE, prob = c(0.55, 0.45))  
  )  
  
  # Train/test split  
  train_idx <- sample(1:n, size = round(n * train_prop))  
  train_data <- data[train_idx, ]  
  test_data <- data[-train_idx, ]  
  
  # empty results  
  results <- list(  

```

```

C_true = NA, Q_true = NA, B_true = NA,
C_lda = NA, Q_lda = NA, B_lda = NA,
C_logit = NA, Q_logit = NA, B_logit = NA
)

#True probabilities for test data (Bayes rule, equal priors)
test_X <- as.matrix(test_data[, c("Age", "Baseline_SBP")])
f0 <- mvtnorm::dmvnorm(test_X, mean = mu0, sigma = Sigma)
f1 <- mvtnorm::dmvnorm(test_X, mean = mu1, sigma = Sigma)
P_true <- f1 / (f0 + f1) # prior probabilities are 0.5 each

Y_test <- test_data$Response
n1 <- sum(Y_test == 1)
n0 <- sum(Y_test == 0)

concordant_true <- Index_C(P_true, Y_test)
results$C_true <- concordant_true / (n1 * n0)
results$Q_true <- 1 + mean(Y_test * log2(P_true) + (1 - Y_test) *
                          log2(1 - P_true))
results$B_true <- 1 - (sum((P_true - Y_test)^2) / length(Y_test))

# LDA
tryCatch({
  lda_model <- lda(Response ~ Treatment + Age + Sex + Baseline_SBP,
                  data = train_data)
  lda_predictions <- predict(lda_model, test_data)
  P_lda <- lda_predictions$posterior[, 2]

  concordant_lda <- Index_C(P_lda, Y_test)
  results$C_lda <- concordant_lda / (n1 * n0)
  results$Q_lda <- 1 + mean(Y_test * log2(P_lda) + (1 - Y_test) *
                          log2(1 - P_lda))
  results$B_lda <- 1 - (sum((P_lda - Y_test)^2) / length(Y_test))
})

# Logistic
tryCatch({
  logit_model <- glm(
    Response ~ Treatment + Age + Sex + Baseline_SBP,
    data = train_data,
    family = binomial
  )

  P_logit <- predict(logit_model, newdata = test_data, type = "response")
  concordant_logit <- Index_C(P_logit, Y_test)

```

```

results$C_logit <- concordant_logit / (n1 * n0)
results$Q_logit <- 1 + mean(Y_test * log2(P_logit) + (1 - Y_test) *
                           log2(1 - P_logit))
results$B_logit <- 1 - (sum((P_logit - Y_test)^2) / length(Y_test))

})

return(
  data.frame(
    maha_dist = maha_dist,
    true_C = results$C_true,
    true_Q = results$Q_true,
    true_B = results$B_true,
    LDA_C = results$C_lda,
    LDA_Q = results$Q_lda,
    LDA_B = results$B_lda,
    LOGIT_C = results$C_logit,
    LOGIT_Q = results$Q_logit,
    LOGIT_B = results$B_logit
  )
)
}

```

1000 times data generated each for different maha distance maha =.5, 1, 1.5, 2

```

all_results <- data.frame()

for (m in maha_values) {
  for (rep in 1:n_replications) {
    res <- run_simulation_maha(n = 1000, rho = 0.5, maha_dist = m, train_prop = 0.7)
    res$Replication <- rep
    res$maha <- m
    all_results <- rbind(all_results, res)
  }
}

```

for each maha distance maha =.5, 1,1.5, 2 we find the mean index and then find which model perform better

```

# Comparison of index averages by maha
Comparison_of_indexes <- all_results %>%
  group_by(maha) %>%
  summarise(

```

```

# C averages
True_C_mean = mean(true_C, na.rm = TRUE),
LDA_C_mean = mean(LDA_C, na.rm = TRUE),
LOGIT_C_mean = mean(LOGIT_C, na.rm = TRUE),

# Q averages
True_Q_mean = mean(true_Q, na.rm = TRUE),
LDA_Q_mean = mean(LDA_Q, na.rm = TRUE),
LOGIT_Q_mean = mean(LOGIT_Q, na.rm = TRUE),

# B averages
True_B_mean = mean(true_B, na.rm = TRUE),
LDA_B_mean = mean(LDA_B, na.rm = TRUE),
LOGIT_B_mean = mean(LOGIT_B, na.rm = TRUE)
)

# Proportion of replications where Logistic regression performs better than LDA
Better_performance_of_logit <- all_results %>%
  group_by(maha) %>%
  summarise(
    Prop_Logit_Better_C = mean(LOGIT_C > LDA_C, na.rm = TRUE),
    Prop_Logit_Better_Q = mean(LOGIT_Q > LDA_Q, na.rm = TRUE),
    Prop_Logit_Better_B = mean(LOGIT_B > LDA_B, na.rm = TRUE)
  )

```

TABLE

```

Comparison_of_indexes %>%
  select(maha, True_C_mean, LOGIT_C_mean, LDA_C_mean) %>%
  gt() %>%
  tab_header(title = "C-index Comparison by Mahalanobis Distance") %>%
  fmt_number(columns = -maha, decimals = 3)

Comparison_of_indexes %>%
  select(maha, True_Q_mean, LOGIT_Q_mean, LDA_Q_mean) %>%
  gt() %>%
  tab_header(title = "Q-index Comparison by Mahalanobis Distance") %>%
  fmt_number(columns = -maha, decimals = 3)

Comparison_of_indexes %>%
  select(maha, True_B_mean, LOGIT_B_mean, LDA_B_mean) %>%
  gt() %>%
  tab_header(title = "B-index Comparison by Mahalanobis Distance") %>%
  fmt_number(columns = -maha, decimals = 3)

```

```
Better_performance_of_logit %>%  
  gt() %>%  
  tab_header(  
    title = "Proportion Logistic Better (by Mahalanobis Distance)"  
  ) %>%  
  fmt_number(columns = -maha, decimals = 3)
```

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